

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION with Open CV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

1. Preprocessing and Feature Detection

Question

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent.

Answer

(a) Analysis of the Issue

Different lighting conditions cause changes in image brightness and contrast. As a result, feature detectors may identify different keypoints in the same scene, leading to inconsistent results.

(b) Problems Identified

- Uneven illumination.
- Low contrast.
- Increased image noise.
- Poor repeatability of detected features.

(c) Solution

Apply preprocessing techniques before feature detection:

- Histogram Equalization or CLAHE for contrast enhancement.
- Gaussian Filter or Median Filter for noise removal.
- Intensity normalization to reduce lighting variations.

(d) Justification

Preprocessing improves image quality, enhances important structures, and ensures that feature detectors identify stable and reliable keypoints.

(e) Conclusion

Preprocessing significantly improves the consistency and accuracy of feature detection under varying lighting conditions.

2. Edge vs Corner Detection

Question

A system uses edge detection but fails to identify key points for matching.

Answer

(a) Analysis

Edge detection identifies only the boundaries of objects, whereas feature matching requires unique and stable points.

(b) Problems

- Edges are not distinctive.
- Many edge pixels appear similar.
- Difficult to establish accurate correspondences.

(c) Solution

Use **Corner Detection** (e.g., Harris Corner Detector or Shi-Tomasi Corner Detector).

(d) Justification

Corners contain intensity changes in multiple directions, making them highly distinctive and easier to match between images.

(e) Conclusion

Corner detection provides reliable keypoints and improves feature matching compared to edge detection alone.

3. Feature Matching under Transformations

Question

Two images of the same object are taken at different scales and orientations.

Answer

(a) Analysis

Traditional feature matching methods are sensitive to changes in scale and rotation, resulting in incorrect matches.

(b) Problems

- Scale variation.
- Rotation variation.
- Changes in viewpoint.
- Matching errors.

(c) Solution

Use **Scale-Invariant Feature Transform (SIFT)** or **SURF**.

(d) Justification

These methods detect keypoints that remain stable under scaling and rotation, making feature matching accurate even when the object appears at different sizes and orientations.

(e) Conclusion

Scale-invariant feature detectors provide robust and reliable matching under geometric transformations.

4. PCA for Feature Optimization

Question

A system extracts many features, leading to high computational cost.

Answer

(a) Analysis

A large number of features increases memory usage, processing time, and redundancy.

(b) Problems

- High-dimensional data.
- Redundant features.
- Increased computational complexity.
- Slower processing.

(c) Solution

Apply **Principal Component Analysis (PCA)** to reduce dimensionality.

(d) Justification

PCA transforms correlated features into a smaller set of principal components while preserving most of the important information. This reduces computation and improves efficiency.

(e) Conclusion

PCA removes redundant information and speeds up feature extraction without significantly affecting accuracy.

5. PCA in Face Recognition

Question

A face recognition system processes high-dimensional image data, resulting in slow performance.

Answer

(a) Analysis

Face images contain thousands of pixel values, making recognition computationally expensive.

(b) Problems

- Large feature vectors.
- Increased memory usage.
- Slow recognition.
- Redundant information.

(c) Solution

Use **Principal Component Analysis (PCA)** (Eigenfaces approach).

(d) Justification

PCA reduces the number of dimensions while retaining the most significant facial features. This decreases storage requirements and speeds up recognition with minimal loss of accuracy.

(e) Conclusion

PCA is an effective dimensionality reduction technique for face recognition, improving efficiency while preserving important facial characteristics.

Code of Conduct:

I V G ABBIRAMY - 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V G ABBIRAMY

RUBICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand